FCM Cygwin Installation:

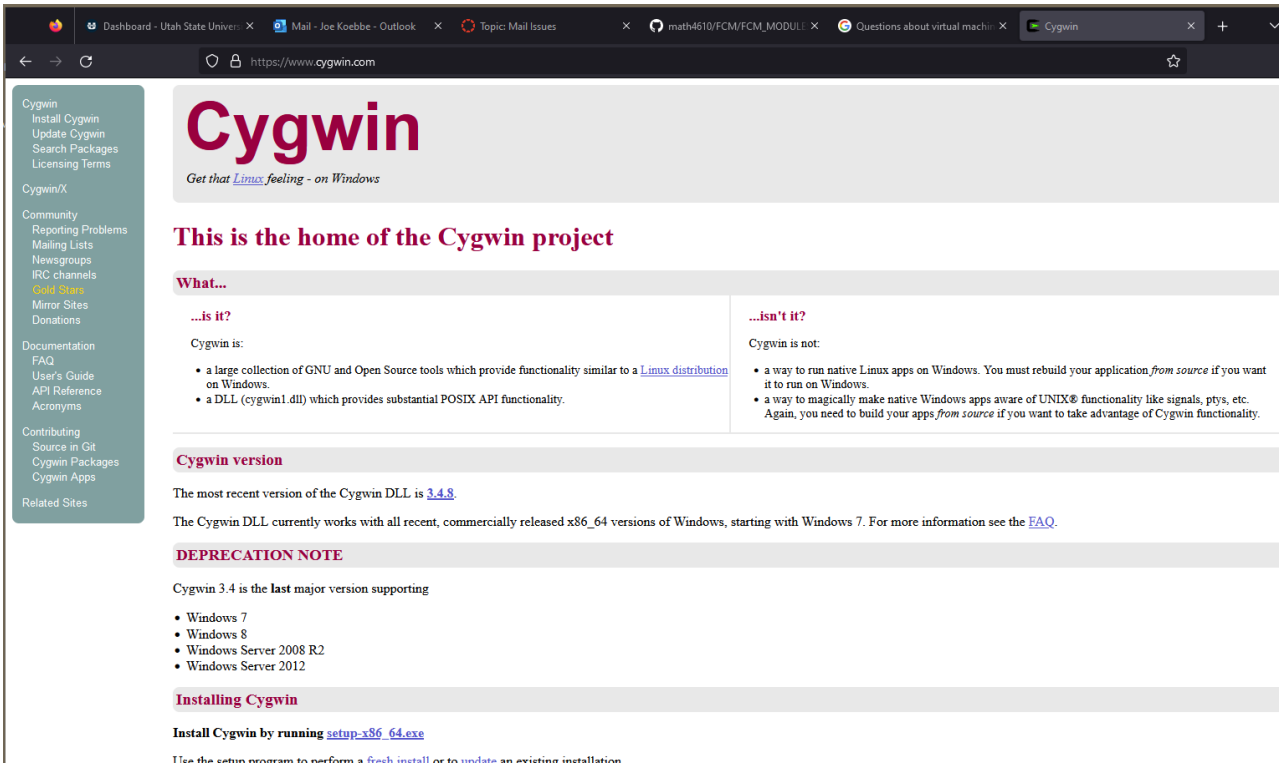
Cygwin is a way to get all the features that you might want in a Linux Environment that is free of charge. Note that if you want a way to operate in a command terminal, Cygwin will allow you to do this while still working on your Windows Desktop. Cygwin allows you to work in parallel to applications with a Windows Desktop. In this lesson, you will learn to install Cygwin for Windows.

Step 1: Finding Cygwin on the Internet

To access Cygwin on the internet, type/copy the following site:

```
http://www.cygwin.org
```

or click on Cygwin.



Then move on to the next step to download the set up script.

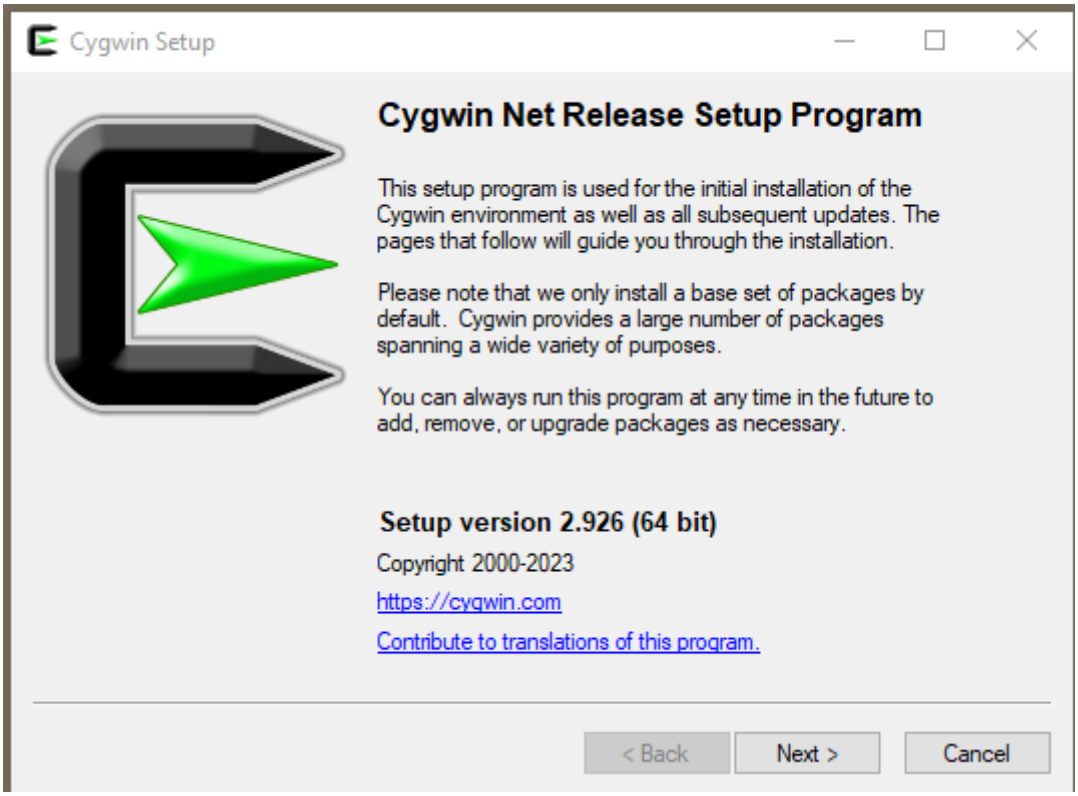
Step 2: Download Setup.exe

When you get the main page for Cygwin, there is a link for "Installing Cygwin". Click on the link setup-x86_64.exe. This will initiate a download of the setup file for Cygwin.

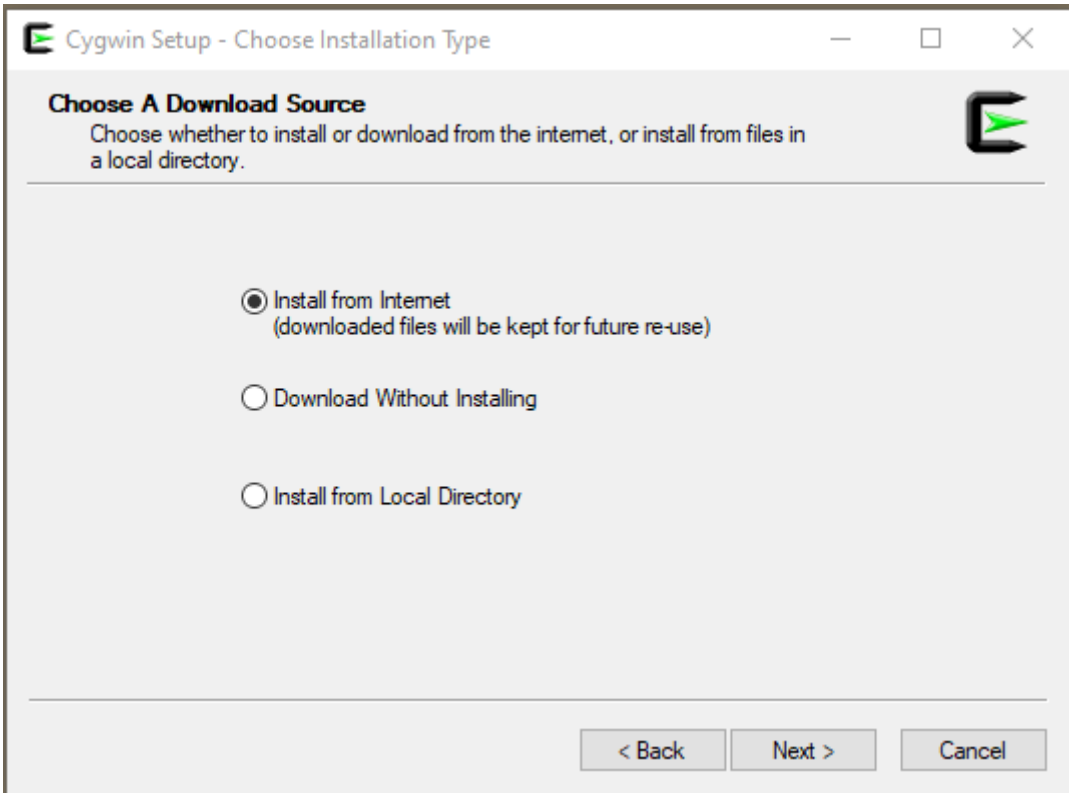
Step 3: Start the Installation:

When going through the installation of Cygwin, you should use the defaults to get most of the things you need. You can add lots more to Cygwin once the initial installation has been completed. There will more discussion of this in the next section. So click on the setup executable that is in you download folder.

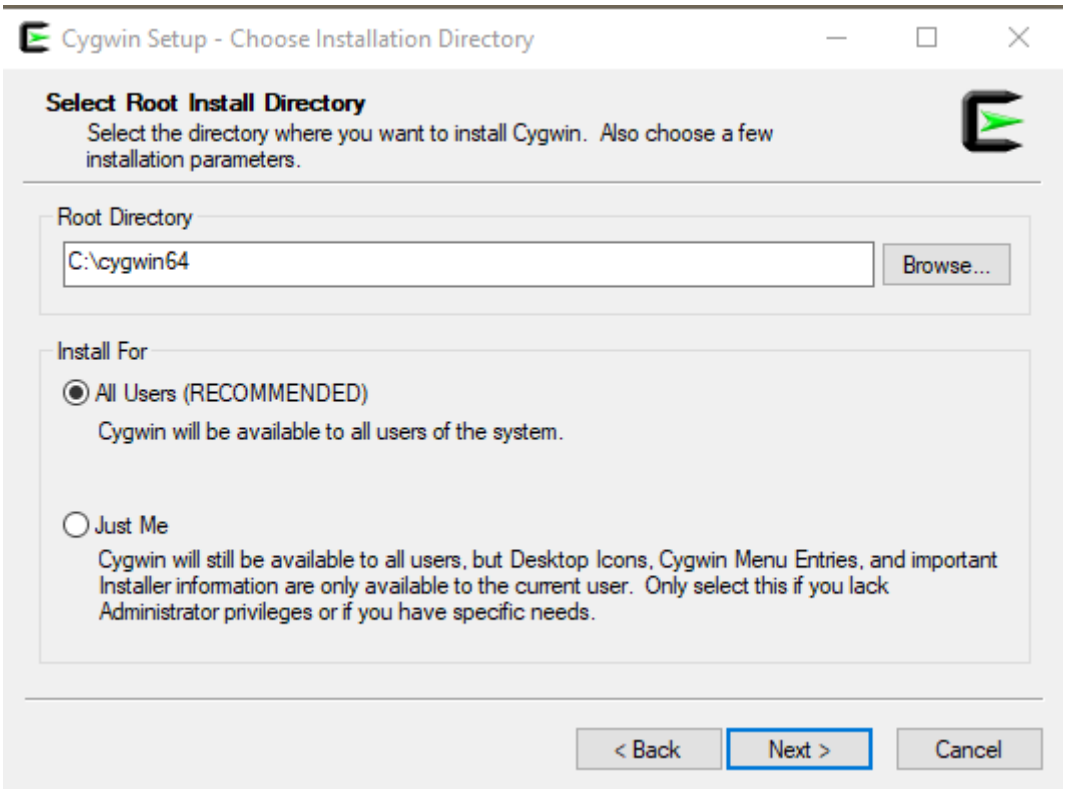
This will start the installation and the dialogs will start appearing. The first is the usual "Do you want to let ...." appears. You can stop the installation if you decide you do not want to continue. If you click on "Yes" the next dialog will look like



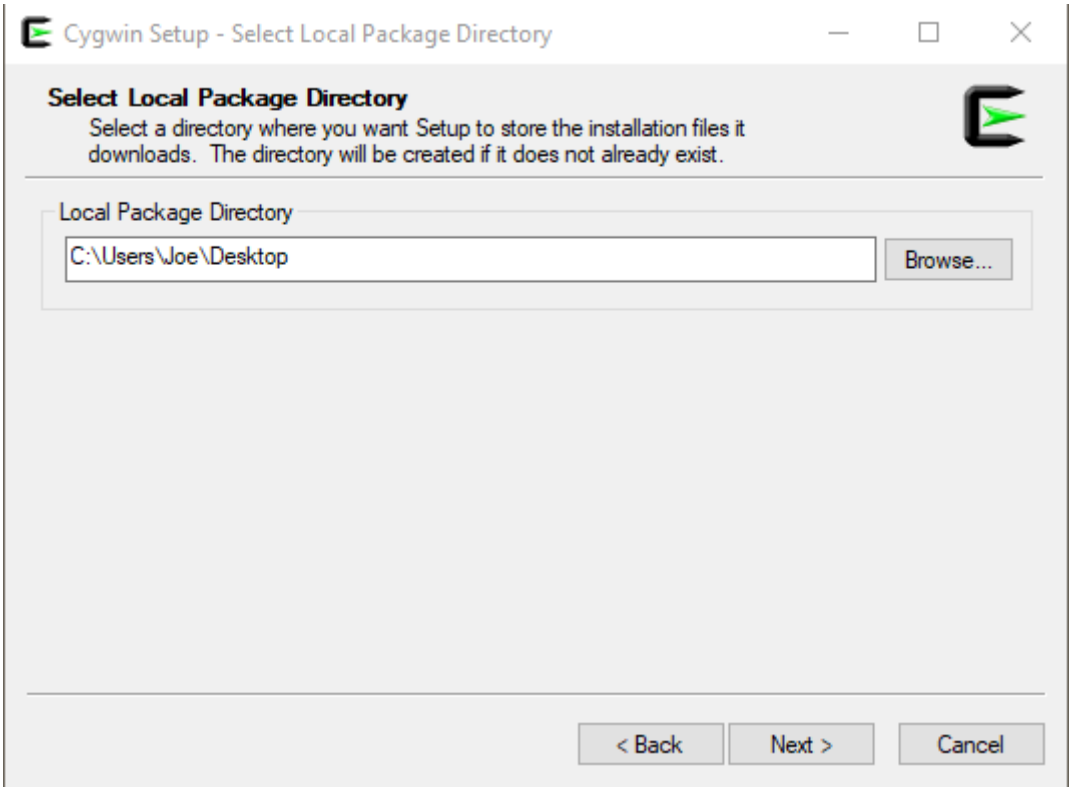
Click on Next to move on. Note that choosing Next on the rest of the prompts will install the default Cygwin distribution.



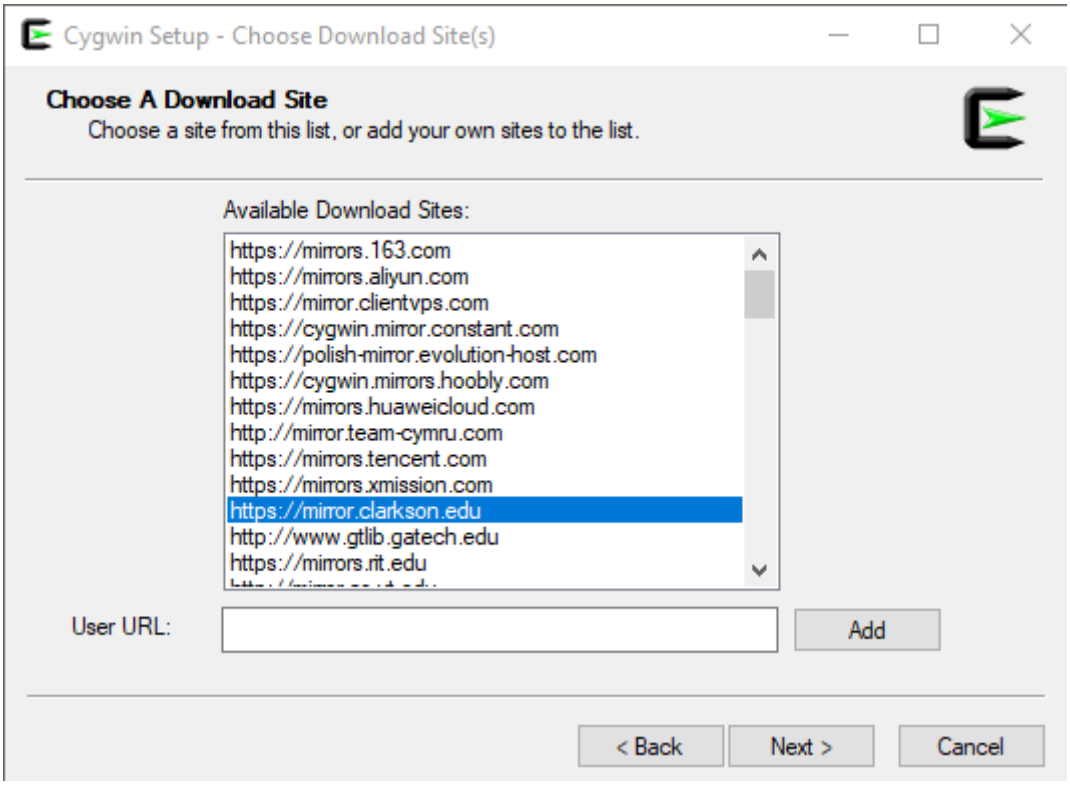
Next ... choose "Next>" to move on. The next prompt will be the for the root installation folder. The dialog will look like



Click "Next>" to move on through the default installation. This results in a dialog like the following.



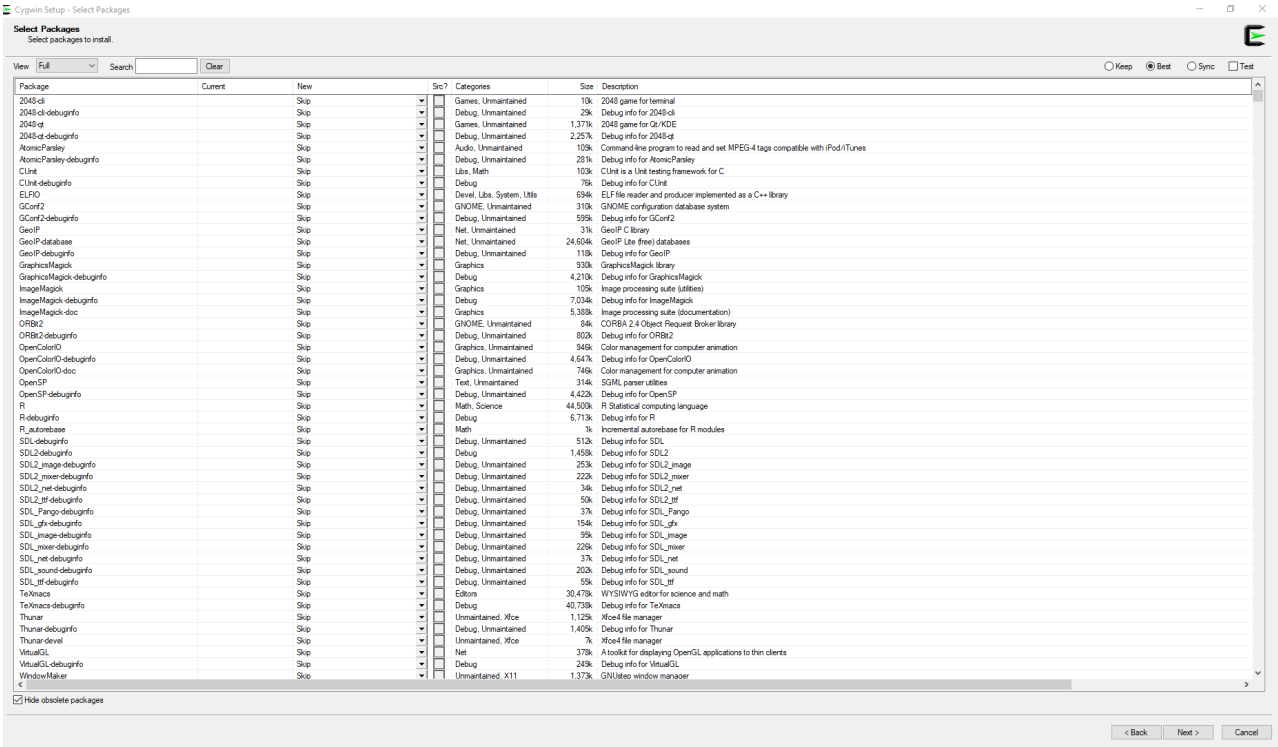
Again, clicking on "Next>" will move along through the default installation. The new dialog will look like the following. A brief dialog requiring no input will flash be and then the following dialog will ask for a location to pick up the installation and modifications we will do.



The site chosen is one that has worked well for the instructor of the course. Now, click "Next>" again. The setup program will work through the info so far and a dialog will popup for a second or two with a progress bar giving you information about what is going on. The default installation will then popup a table of packages to install.

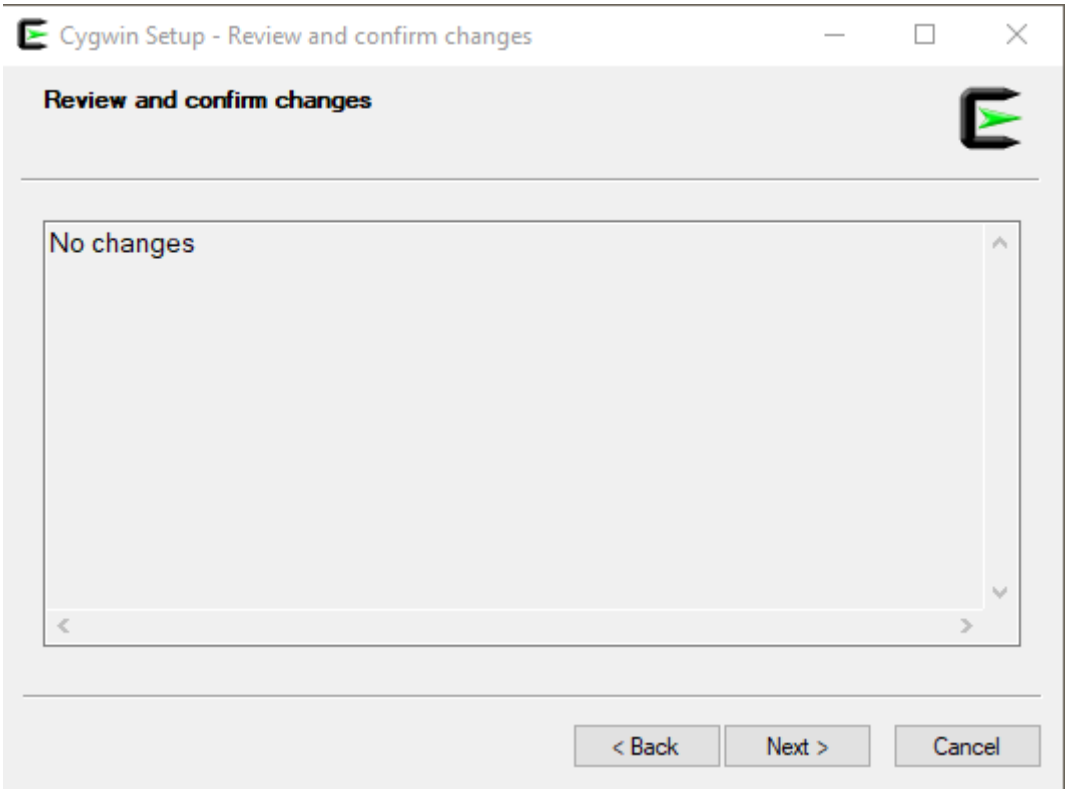
IMPORTANT NOTE Do not start with the Full distribution unless you have a lot of disk space and a lot of time and patience. You can always add packages to the distribution in Cygwin. There is an example in the last section of this module.

So, we see something like

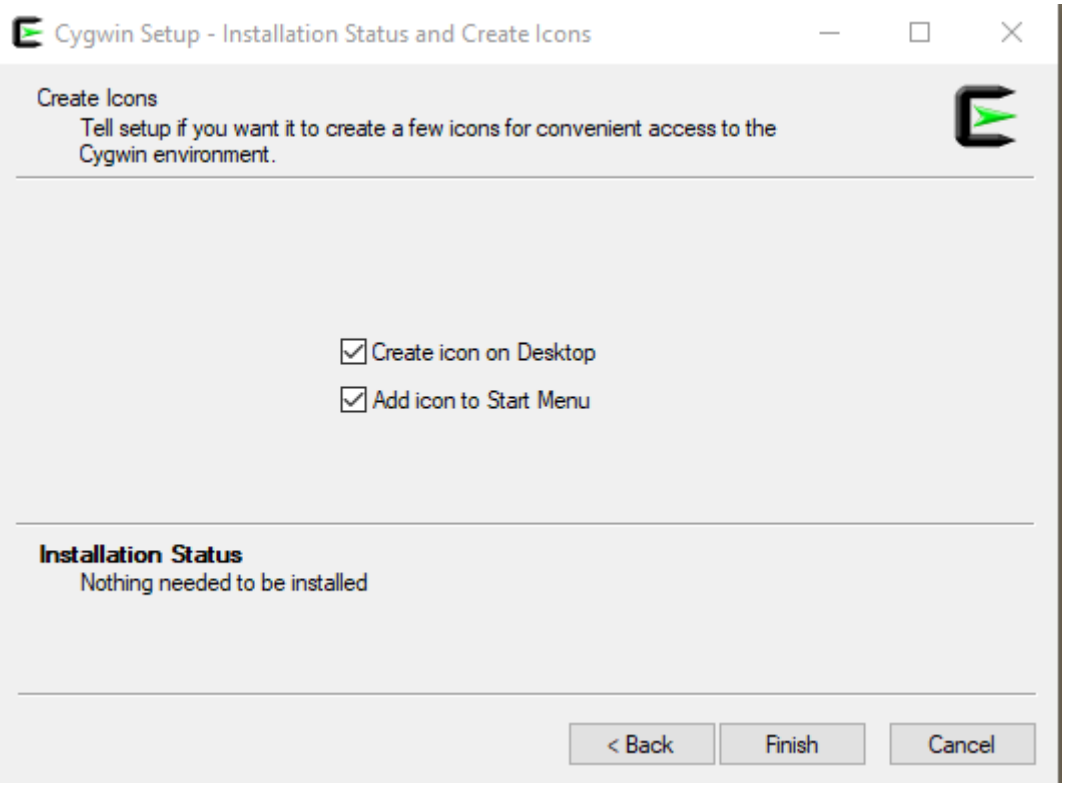


Package	Current	New	Src?	Categories	Size	Description
2048-cli		Skip		Games, Unmaintained	10k	2048 game for terminal
2048-cli-debuginfo		Skip		Debug, Unmaintained	29k	Debug info for 2048-cli
2048-qt		Skip		Games, Unmaintained	1,371k	2048 game for Qt/KDE
2048-qt-debuginfo		Skip		Debug, Unmaintained	2,257k	Debug info for 2048-qt
AtomicParsley		Skip		Audio, Unmaintained	109k	Command-line program to read and set MPEG-4 tags compatible with iPod/iTunes
AtomicParsley-debuginfo		Skip		Debug, Unmaintained	281k	Debug info for AtomicParsley
CUnit		Skip		Libs, Math	103k	CUnit is a Unit testing framework for C
CUnit-debuginfo		Skip		Debug	76k	Debug info for CUnit
ELFIO		Skip		Devel, Libs, System, Utils	694k	ELF file reader and producer implemented as a C++ library
GConf2		Skip		GNOME, Unmaintained	310k	GNOME configuration database system
GConf2-debuginfo		Skip		Debug, Unmaintained	595k	Debug info for GConf2
GeoIP		Skip		Net, Unmaintained	31k	GeoIP C library
GeoIP-database		Skip		Net, Unmaintained	24,604k	GeoIP Lite (free) databases
GeoIP-debuginfo		Skip		Debug, Unmaintained	118k	Debug info for GeoIP
GraphicsMagick		Skip		Graphics	930k	GraphicsMagick library
GraphicsMagick-debuginfo		Skip		Debug	4,210k	Debug info for GraphicsMagick
ImageMagick		Skip		Graphics	105k	Image processing suite (utilities)
ImageMagick-debuginfo		Skip		Debug	7,034k	Debug info for ImageMagick
ImageMagick-doc		Skip		Graphics	5,388k	Image processing suite (documentation)
ORBit2		Skip		GNOME, Unmaintained	84k	CORBA 2.4 Object Request Broker library
ORBit2-debuginfo		Skip		Debug, Unmaintained	802k	Debug info for ORBit2
OpenColorIO		Skip		Graphics, Unmaintained	948k	Color management for computer animation
OpenColorIO-debuginfo		Skip		Debug, Unmaintained	4,647k	Debug info for OpenColorIO
OpenColorIO-doc		Skip		Graphics, Unmaintained	746k	Color management for computer animation
OpenSP		Skip		Text, Unmaintained	314k	SGML parser utilities
OpenSP-debuginfo		Skip		Debug, Unmaintained	4,422k	Debug info for OpenSP
R		Skip		Math, Science	44,500k	R Statistical computing language
R-debuginfo		Skip		Debug	6,713k	Debug info for R
R_autorebase		Skip		Math	1k	Incremental autorebase for R modules
SDL-debuginfo		Skip		Debug, Unmaintained	512k	Debug info for SDL
SDL2-debuginfo		Skip		Debug	1,458k	Debug info for SDL2
SDL2_image-debuginfo		Skip		Debug, Unmaintained	253k	Debug info for SDL2_image
SDL2_mixer-debuginfo		Skip		Debug, Unmaintained	222k	Debug info for SDL2_mixer
SDL2_net-debuginfo		Skip		Debug, Unmaintained	34k	Debug info for SDL2_net
SDL2_ttf-debuginfo		Skip		Debug, Unmaintained	50k	Debug info for SDL2_ttf
SDL_Pango-debuginfo		Skip		Debug, Unmaintained	37k	Debug info for SDL_Pango
SDL_gfx-debuginfo		Skip		Debug, Unmaintained	154k	Debug info for SDL_gfx
SDL_image-debuginfo		Skip		Debug, Unmaintained	95k	Debug info for SDL_image
SDL_mixer-debuginfo		Skip		Debug, Unmaintained	226k	Debug info for SDL_mixer
SDL_net-debuginfo		Skip		Debug, Unmaintained	37k	Debug info for SDL_net
SDL_sound-debuginfo		Skip		Debug, Unmaintained	202k	Debug info for SDL_sound
SDL_ttf-debuginfo		Skip		Debug, Unmaintained	55k	Debug info for SDL_ttf
TeXmacs		Skip		Editors	30,478k	WYSIWYG editor for science and math
TeXmacs-debuginfo		Skip		Debug	40,738k	Debug info for TeXmacs
Thunar		Skip		Unmaintained, Xfce	1,125k	Xfce4 file manager
Thunar-debuginfo		Skip		Debug, Unmaintained	1,405k	Debug info for Thunar
Thunar-devel		Skip		Unmaintained, Xfce	7k	Xfce4 file manager
VirtualGL		Skip		Net	378k	A toolkit for displaying OpenGL applications to thin clients
VirtualGL-debuginfo		Skip		Debug	249k	Debug info for VirtualGL
WindowMaker		Skip		Unmaintained, X11	1,373k	GNUstep window manager

We are almost to the last dialog. Clicking on "Next>"



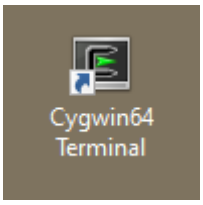
Finally, a dialog will popup to see if a couple icons can be created. That will actually help us get to a terminal and a way to work with a Linux OS.



Click on "Finish" to complete the installation. You are done. If you have clicked the check box(es) you will have a couple of ways to a terminal.

Bringing Up a Cygwin Terminal.

If you installed the icons that are included in the last step of installation, the following icon should appear somewhere on your desktop.



Double clicking on the icon will open a terminal, which is what we want. So we see a window that looks like the following.

Note that once this is done, we can get at a version of Ubuntu Linux that will allow us some serious control over both the Cygwin environment and Windows at the same time. For now we can test our knowledge of the OS.

Updating the Cygwin Distribution

To add packages or modify/upgrade packages in Cygwin, you can run the setup_x86_64.exe again. In fact, you can rerun the setup program as many times as you want. The software underlying Cygwin is being modified continuously. So, rerunning the setup program every week or so is a good idea. If makes getting the latest revision of the software relatively easy.

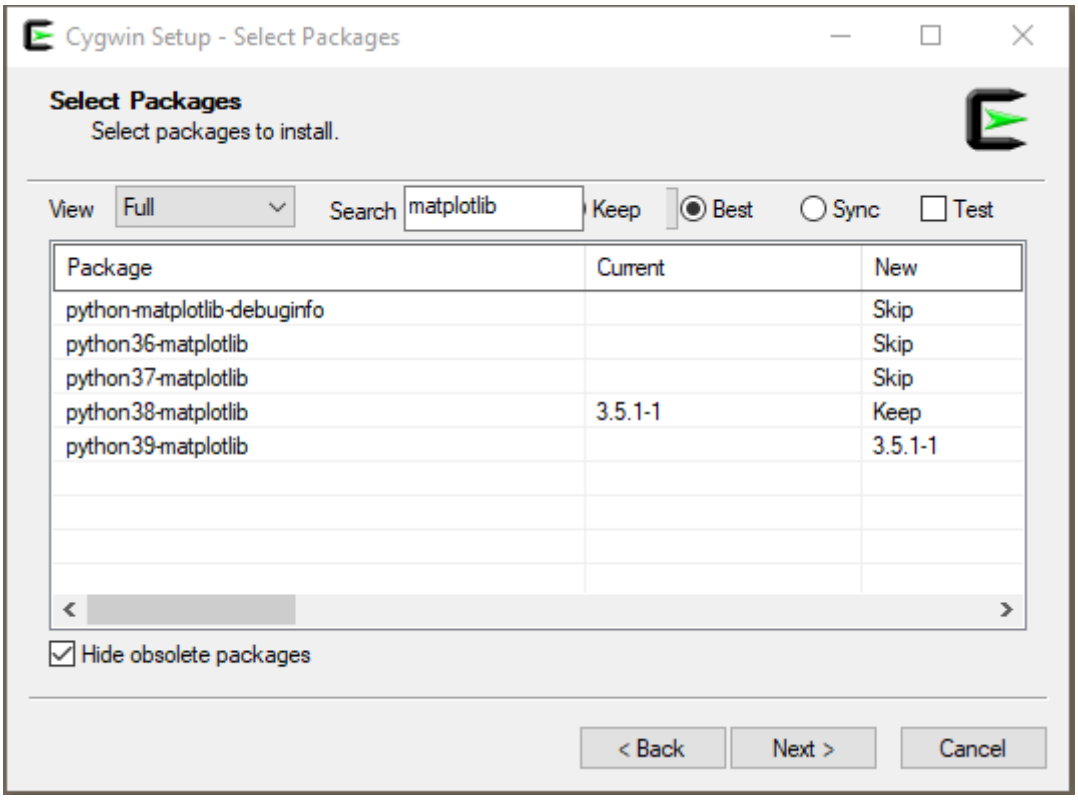
All you need to do is click on every default and when the table of packages shows up, one of two things will happen. Either there will be a list of packages that need updating or the table will show no entries. In the first case, the, clicking on "Next" and "Finish" will complete the update. If there is nothing to update, you do the same thing with no package updates.

The following is a very specific example of how to include a package in Cygwin.

Installing Packages Not in the Default Installation.

If there are packages that are not installed automatically by Cygwin, you can use the setup program to identify all of the software you need for the additional installation along

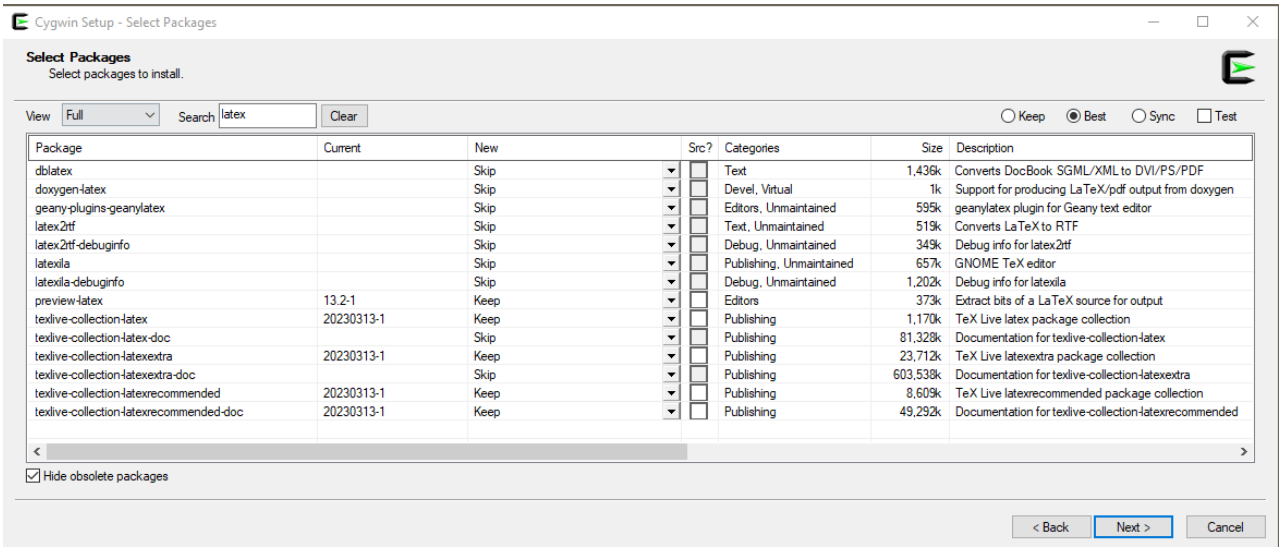
with completing the installation of the software. An example of the process of installing another package, the following window shows the step just before the installation starts.



The effort in this case is to install a package, python.39-matplotlib, to work with in the Cygwin environment. All you need to do is to click on the "Next>" button to move on. The setup app should be run occasionally to make sure the latest updates to the Cygwin environment.

A Few Remarks

- Every once in awhile you should download the latest version of the setup executable. Changes to this executable are less frequent than the packages. So this will not be a lot of additional work to maintain the Cygwin environment.
- The packages related to latex, including executables, style files, and fonts can be loaded into the Cygwin environment. Notice in the following screen shot that the View is "Full" meaning that this out of all possible packages. In the Search field the string "latex" has been entered. So, the interface will list only the packages that have "latex" in the package name. The rest of the work is in determining what the correct package is. The Search file can be used to narrow down what you really need. Note also that the size of the download is pretty large if the fonts are all installed.
- When trying to install packages you want, it is a good idea to find what you really need before you let the modified installation start up.
- If things really mess up it is very easy to start over by getting rid of some folders. However, it will take a bit of time to get there.



Package	Current	New	Src?	Categories	Size	Description
dblatex		Skip		Text	1,436k	Converts DocBook SGML/XML to DVI/PS/PDF
doxygen-latex		Skip		Devel, Virtual	1k	Support for producing LaTeX/pdf output from doxygen
geany-plugins-geanylatex		Skip		Editors, Unmaintained	595k	geanylatex plugin for Geany text editor
latex2rtf		Skip		Text, Unmaintained	519k	Converts LaTeX to RTF
latex2rtf-debuginfo		Skip		Debug, Unmaintained	349k	Debug info for latex2rtf
latexila		Skip		Publishing, Unmaintained	657k	GNOME TeX editor
latexila-debuginfo		Skip		Debug, Unmaintained	1,202k	Debug info for latexila
preview-latex	13.2-1	Keep		Editors	373k	Extract bits of a LaTeX source for output
texlive-collection-latex	20230313-1	Keep		Publishing	1,170k	TeX Live latex package collection
texlive-collection-latex-doc		Skip		Publishing	81,328k	Documentation for texlive-collection-latex
texlive-collection-latexextra	20230313-1	Keep		Publishing	23,712k	TeX Live latexextra package collection
texlive-collection-latexextra-doc		Skip		Publishing	603,538k	Documentation for texlive-collection-latexextra
texlive-collection-latexrecommended	20230313-1	Keep		Publishing	8,609k	TeX Live latexrecommended package collection
texlive-collection-latexrecommended-doc	20230313-1	Keep		Publishing	49,292k	Documentation for texlive-collection-latexrecommended

Programming Example: Edit, Compile, Run, Repeat

In this section we will write a simple program in a terminal using a text editor called, vim, compile the code in the file using an appropriate compiler (c, gfortran, or java), and then attempt to execute the program. Before this, we will need to agree on a shell that will be used for the work.

> **Definition** In the realm of computers, a shell is a computer program that allows a person to access services provided by an operating system. A computer user can use a command line interface that will take commands from a user and translate these into responses that effect the state of the computer. It is called a shell.

We will be able to use any of the commands available in a shell to change what the machine creates, modifies, kills, and so on. In the next example, we will write a small program to show how to use a terminal in the process. Let's program up the infamous "hello world" example in the C-programming language. We will do this over a few steps and we will use several commands that Cygwin has available. The commands we will use are:

1. **tcsh** - this is a shell that is available in Cygwin, although may have to install the package using the process discussed above. Note that the most common starting point for a terminal is **bash**. The instructor in the class has opted into **tcsh** due to learning C at the same time and these two behave similarly.
2. **vim** - this is a terminal based editor and is a bit cryptic. However, this usually available on any Linux/Unix computer.
3. "gcc" - this is a C compiler that can be used to compile a text file into an executable.
4. If the compilation set is successful, an executable/command will be produced in the process. This command will be excuted.

Launch a Terminal to Work in

If you have Cygwin installed, you should be able to start by double clicking on the Cygwin icon as before. The following terminal should appear.

Then move to the next step.

Step 1. Make sure the commands are installed.

The programs we will need are the following:

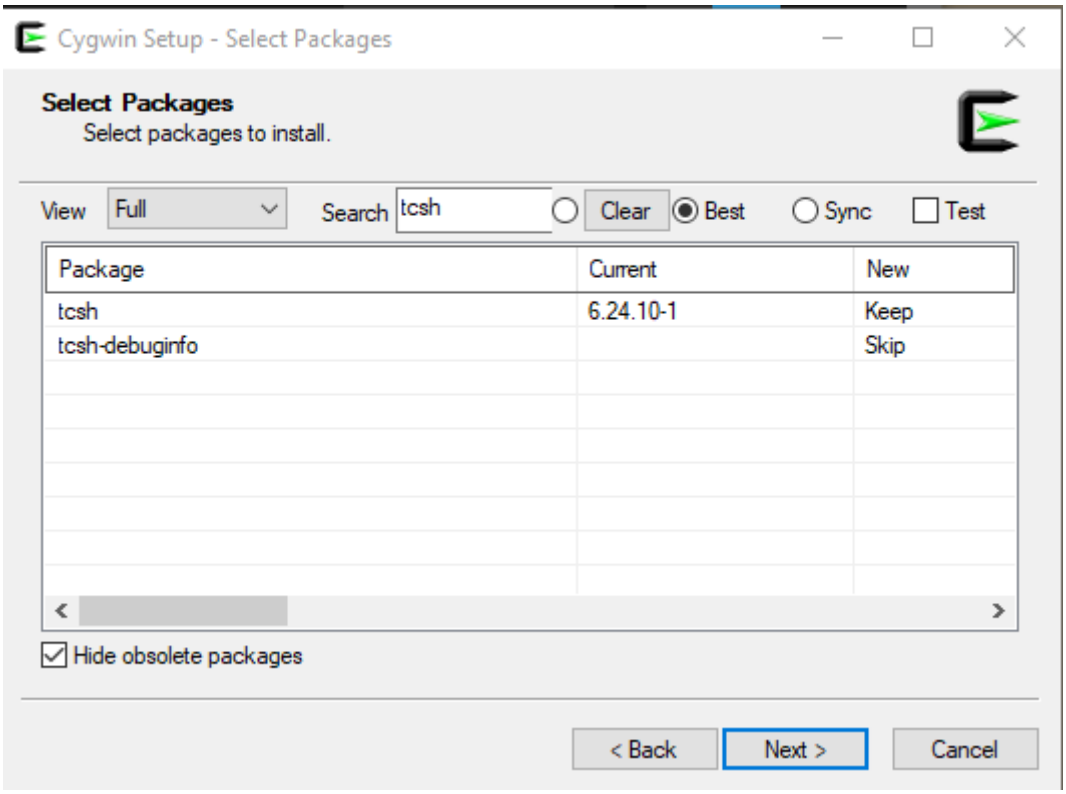
- tcsh - the shell to use in the example,
- vim - the text file editor
- gcc

The end command will combine these commands. To test if the commands are available enter the commands as shown in the next figure.

```
44 Joe: which tcsh
/bin/tcsh
45 Joe: which vim
/bin/vim
46 Joe: which gcc
/bin/gcc
47 Joe:
```

If no warnings errors occur about "command not found." you can move to the next step. If the commands generate a warning or error, you will need to go back to the setup program to install.

Let's assume that the **tcsh** command is not available. Then, run the Cygwin to the table showing the packages. From this, select Full in the drop down menu and Search for "tcsh". If all goes well the following will appear in your installation setup sequence.

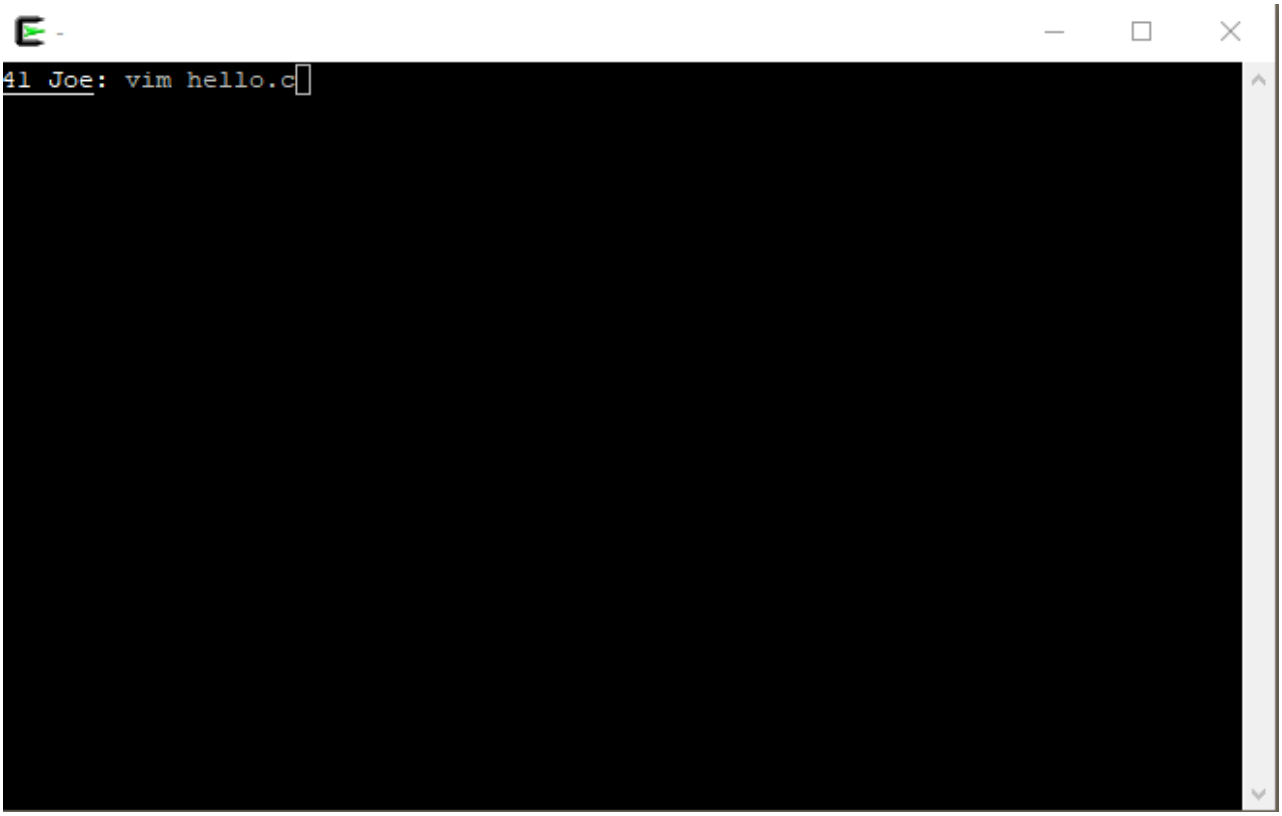


In this case, the tcsh package has been installed. So there should be no problems with the command.

Note that you may need to do this with both **vim** and **gcc** commands to finish this off. However, if the **which** command is not producing warnings we can move on.

Step 2. Editing the Text File

Now we need to create the program file. The next command is to invoke the editor, **vim**. This can be seen in a terminal as follow.



The result is a terminal that looks like the following:

```
"hello.c" [New]                                          0,0-1         All
```

At this point you see a blank file and a line along the bottom indicating that the file is [New] and the name of the file is "hello.c". To create the file, you can type a lower case "a" to add text to the file. and add the text so that the terminal looks like the following.

```
#include <stdio.h>

int main()
{
  printf("Hello World!\n");
}
-- INSERT --                                             6,2           All
```

When you type the "a" at the beginning the editor is put into an append mode which will append characters as are typed at the key board. When there is a new line, you can type a "return/enter" to move to the next line. If you get the window to look exactly like above,

type and ESC on the terminal and you will be taken out of append ode. The window should look like

```
#include <stdio.h>

int main()
{
  printf("Hello World!\n");
}
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
                                                    6,1           All
```

Once the file is no longer in append mode, you can type a ":x" to exit the editor and save your work. The terminal will look like

```
#include <stdio.h>

int main()
{
  printf("Hello World!\n");
}
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
:x
```

and then the terminal will look like the following.

```
41 Joe: vim hello.c
42 Joe:
```

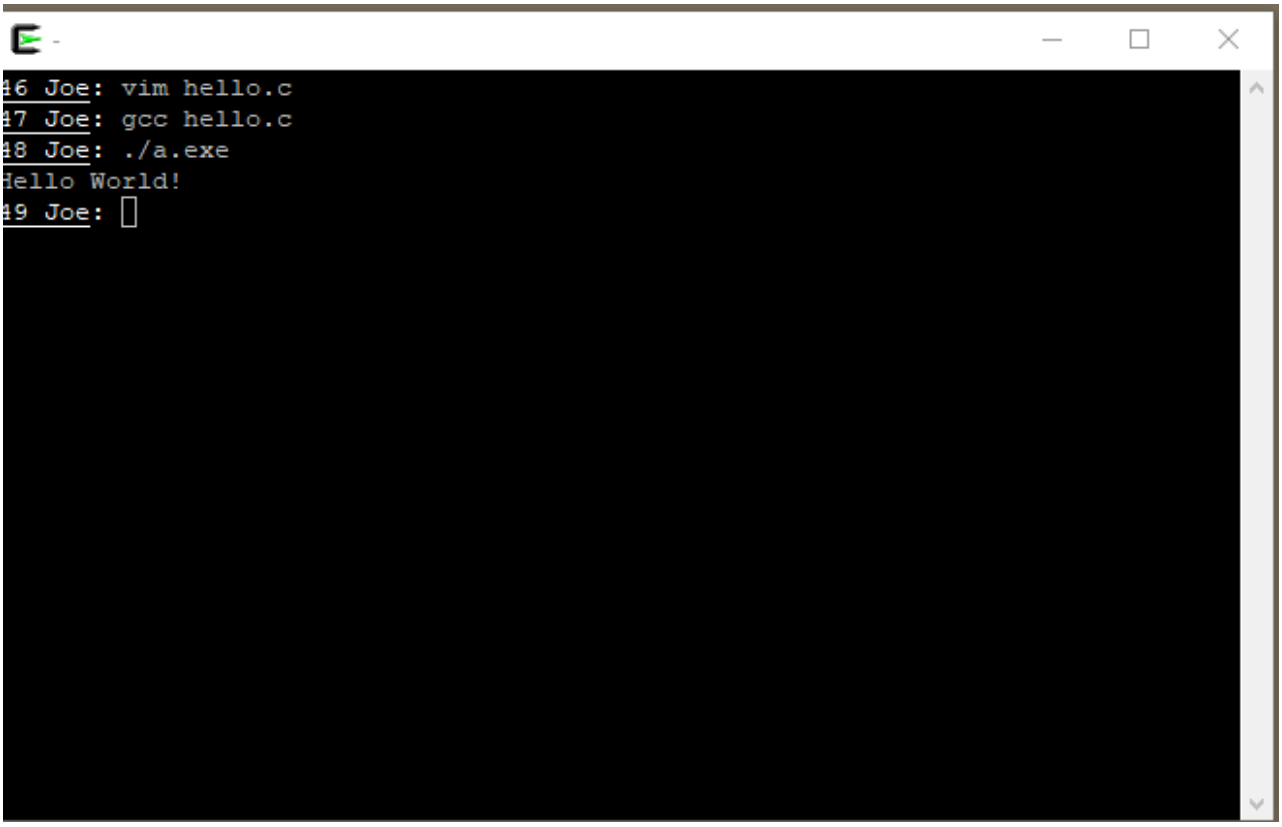
If all has gone smoothly, the text file, "hello.c" should be in folder. You can use the **ls** command at the prompt to see this. We are ready to move on to the next step.

Step 3. Compiling the Program.

If we have made sure that **gcc** is installed, we are ready to compile. To create a simple executable, make the terminal look like the following and hit Enter to execute the command. The terminal will look like the following.

```
46 Joe: vim hello.c
47 Joe: gcc hello.c
48 Joe:
```

If there are no errors and the like, we are ready to execute the executable version of the program we have created. If no name is given to the executable file, the default name is **a.exe** or sometimes **a.out** depending on your computer. After you type in the command **a.exe** at the command prompt the final display will be the following terminal.



We are done with the example and have created a text file to encode our program, compiled the program, and then executed the program to see some output from a computer.

Summary:

Cygwin can be use as a powerful computer platform. There are a lot of pros and cons to this. However, if you need a terminal for testing what you are doing on a Windows box and you need flexibility, Cygwin is one way to go.

Questions and Problems.

Question 1 How does Cygwin differ from other Linux/Unix operating systems?

Question 2 What are the two main shell scripting environments?

Question 3 Write, compiler and run a "Hello World!" application in Python and use the python interpreter to run your code.

References:

- Wikipedia definition of an operating system shell with explanations.

- Introduction to Cygwin